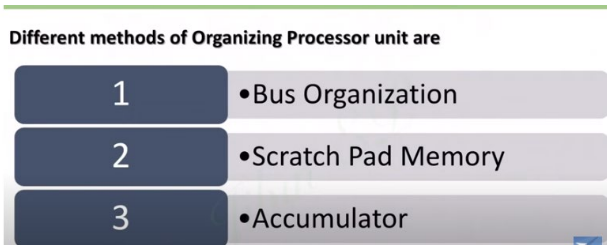
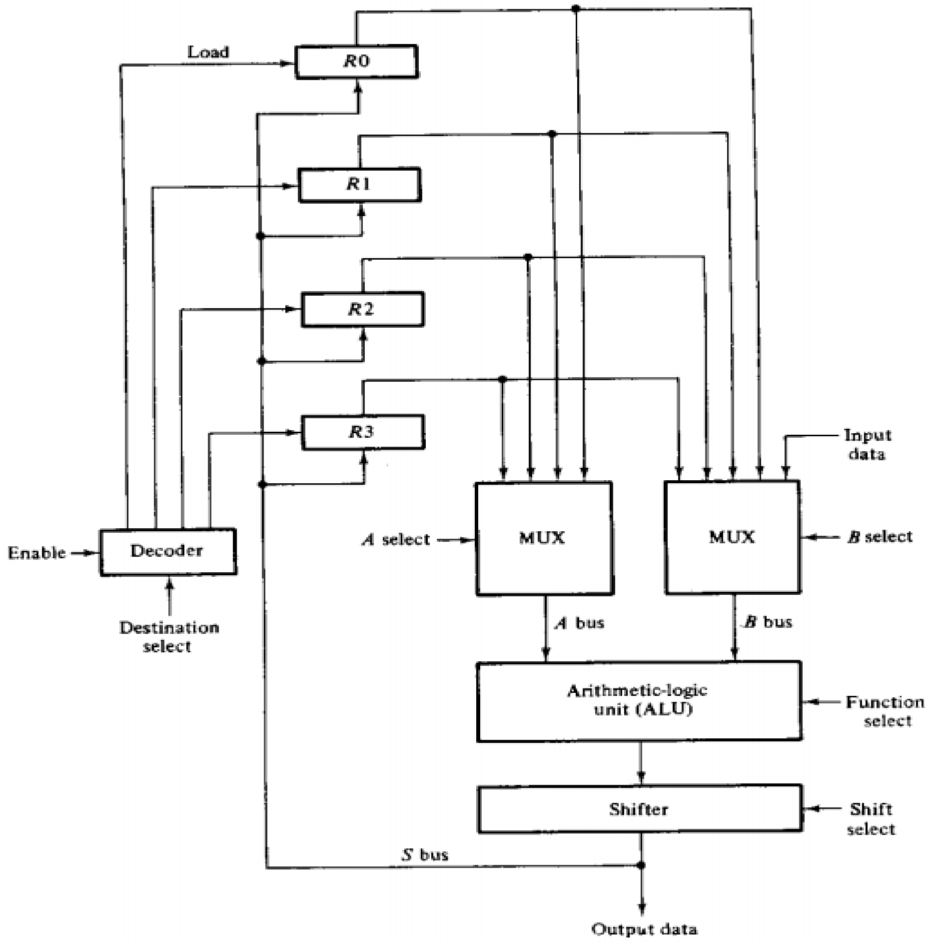
**PROCESSOR ORGANIZATION**

- Processor part -data path of the CPU (Because processor forms the paths for the data transfers between the registers in the unit
- Data paths by buses and other common lines
- Control gates that formulate the given path are - multiplexers and decoders whose selection lines specify the required path.




**<u>Bus Organization(4 processor registers)</u>**



- Register connected to two MUX to form input buses A and B.
- The selection lines of each MUX select one register for the particular bus.
- The A and B buses are applied to a common ALU.
- The function selected in the ALU determines the particular operation that is to be performed.
- The shift micro-operations are implemented in the shifter.
- The result of the micro-operation goes through the output bus S into the inputs of all registers.
- The destination register that receives the information from the output bus is selected by a decoder.
- When enabled, this decoder activates one of the register load inputs to provide a transfer path between the data on the S bus and the inputs of the selected destination register.
- The output bus S provides the terminals for transferring data to an external destination.
- One input of MUX A or B can receive data from the outside
- The control unit that supervises the processor bus system directs the information flow through the ALU by selecting the various components in the unit.

**Example**

To perform the microoperation:

$R1 \leftarrow R2 + R3$

- The control must provide binary selection variables to the following selector inputs:

1. ***MUX A selector: to place the contents of R2 onto bus A.***
2. ***MUX B selector: to place the contents of R3 onto bus B.***
3. ***ALU function selector: to provide the arithmetic operation A + B.***
4. ***Shift selector: for direct transfer from the output of the ALU onto output bus S (no shift).***
5. ***Decoder destination selector: to transfer the contents of bus S into R 1***

## Scratchpad Memory

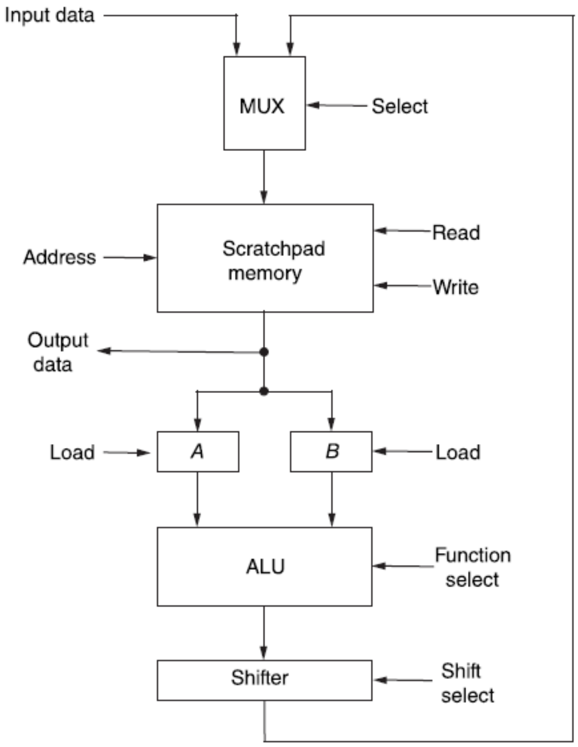
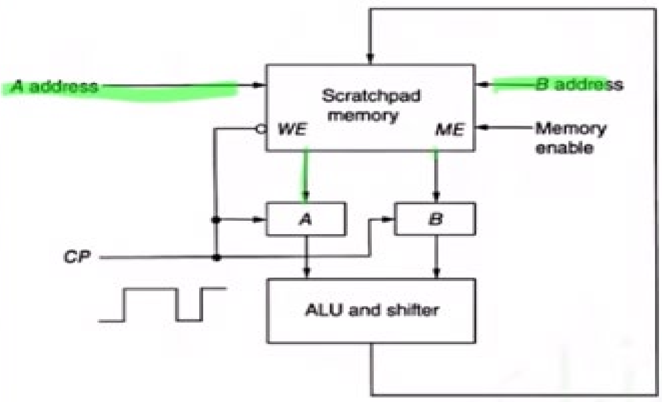



**Figure 9-2** Processor unit employing a scratchpad memory

## Scratchpad Memory

- The registers in a processor unit can be enclosed within a small memory unit.
- When included in a processor unit, a small memory is sometimes called a *scratchpad* memory.
- The use of a small memory is a cheaper alternative to connecting processor registers through a bus system.
- The difference between the two systems is the manner in which information is selected for transfer into the ALU.
- Ina bus system, the information transfer is selected by the multiplexers that form the buses.
- On the other hand, a single register in a group of registers organized as a small memory must be selected by means of an address to the memory unit.
- The information stored in the scratchpad memory would normally come from the main memory by means of instructions in the program.

## 2b-TWO PORT MEMORY




- 2 Separate Address Lines to Select two Words of Memory simultaneously
- 2 Source Registers read at same time
- If the destination register is same as one of the source register then entire microoperation can be done within one clock period

- When Enabled by Memory Enable (ME) the destination will be set as B

- The Clock Input Controls the Memory Read and Write Operation through the Write Enable Input (WE)

**Accumulator Register** –

- Some processor units separate one register from all others and call it an *accumulator* register, abbreviated *AC* or *A* register.
- The accumulator register in a processor unit is a multipurpose register capable of performing not only the add microoperation, but many other microoperations as well.
- Infact, the gates associated with an accumulator register provide all the digital functions found in an ALU.

**Accumulator Register -** for short-term, intermediate storage of arithmetic and logic data in a computer's CPU

- To find the Sum of two numbers stored in processor registers:
- T1: A ← 0 Clear A
- T2: A ← A + R1 Transfer R1 to A
- T3: A ← A + R2 Add R2 to A